

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

**Q. 1 Solve Any Two of the following.**

[12]

**A) Define the following terms:**

Remember

- i) Classification of Data Structure
- ii) Abstract Data Types
- iii) Transpose of Matrix

**B) What is the Sparse Matrix. Write a C program to convert a sparse matrix, an input provided by a user into its triplet representation.**

Understand

**C) Consider the following 4-digit employee numbers: 3205 and 7148. Find 2-digit hash address of each number using a) division method ( $m = 97$ ); b) mid-square method; c) folding method without reversing.**

Application

**Q.2 Attempt the following questions.**

[12]

**A) Define Queue. Write an algorithm to perform an Enqueue and Dequeue operations on the Queue.**

Understand

**B) Write an algorithm to convert Infix expression to Postfix expression. Consider the following Infix expression Q and translate Q into its equivalent Postfix expression P.**

Application

$$Q: A + ( B * C - ( D / E \uparrow F ) * G ) * H$$

**Q. 3 Solve Any Two of the following.**

[12]

**A) Explain the following concepts:**

Understand

- a) Garbage Collection
- b) Dynamic Memory Allocation

**B) Write a pseudo code for the performing following operations in the Single Link List:**

Understand

- a) Insert the elements at the end
- b) Delete the element from the beginning

**C) Write a pseudo code for the performing following operations in the Circular Link List:**

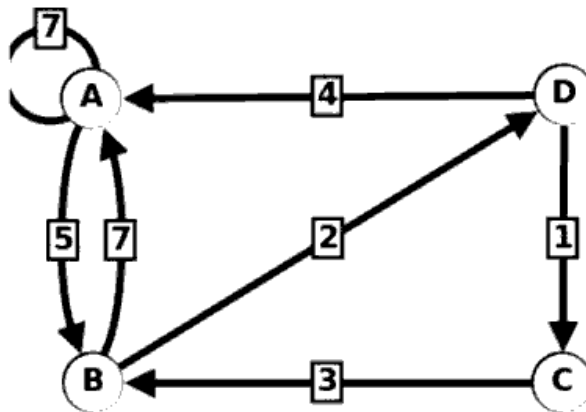
Understand

- a) Insert the elements at the beginning
- b) Delete the element from the end

**Q.4 Attempt the following questions.**

[12]

- A) Consider the following weighted graph G. Find the shortest path between the nodes using Warshall's Algorithm.



- B) Define the the following terms:

Application

- i) Siblings      ii) Leaf Node      iii) Ancestor of Node

Following numbers are inserted into an empty binary search tree. Find the final tree T.

25, 20, 10, 36, 22, 5, 1, 8, 30, 12, 15, 40, 28, 38, 48, 45, 50

**Q. 5 Attempt the following questions.**

[12]

- A) Write a pseudo code for Linear Searching.

Understand

- B) Consider the following array consisting of 8 elements. Use insertion sort to arrange the elements in the Ascending order.

77, 33, 44, 11, 88, 22, 66, 55

\*\*\* End \*\*\*